DNP3

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

DNP3 is the dominant SCADA protocol in North American power distribution and water utility systems, designed for the latency tolerance, event-driven reporting, and data integrity requirements of field device communication over unreliable WAN links. This course covers the three-layer protocol stack, typed data objects, function codes including Select-Before-Operate, unsolicited reporting configuration, and Secure Authentication version 5.

PREREQUISITES

Basic TCP/IP networking concepts. Completion of the Modbus course is recommended.

COURSE TOPICS

1. **Introduction:** Protocol origins, design goals over Modbus, CRC integrity, event-driven reporting, and the typed object model for power utility SCADA.
2. **Protocol Layers:** Application, Transport, and Data Link layers; DNP3's internal three-layer stack and operation on raw serial.
3. **Data Link Layer:** Sync bytes, addressing, control byte, CRC-16 on every 16-byte block, and primary/secondary station roles.
4. **Application Layer:** APDU structure, function codes, object headers, sequence numbers, fragmentation, and multi-fragment exchange.
5. **Data Objects:** Typed object groups and variations — Analog Input Group 30, Binary Input Group 1, quality bytes, and real-deployment object sets.
6. **Function Codes:** Read, Write, Direct Operate, Select-Before-Operate, Freeze, Enable/Disable Unsolicited; SBO two-message sequence and timeout.
7. **Unsolicited Responses:** Class-based event assignment, event buffer sizing, unsolicited confirmation, and master-offline behavior.
8. **Secure Authentication v5:** HMAC challenge/response, pre-shared key management, aggressive mode, supported algorithms, and NERC CIP alignment.
9. **Troubleshooting:** Event buffer overflow, SBO confirm timeout, application layer confirm timeout, and TCP session management on flapping WAN links.
10. **Protocol Lab:** Outstation commissioning, unsolicited reporting setup, SAv5 key management, tshark filtering, and event buffer overflow diagnosis.

LEARNING OUTCOMES

- Configure an outstation point database, event class assignments, and unsolicited reporting from scratch.
- Trace and verify a Select-Before-Operate sequence in a Wireshark protocol capture.
- Configure Secure Authentication v5 key management on a DNP3 master/outstation link.
- Diagnose event buffer overflow, SBO timeout, and application layer confirm timeout failures.
- Write tshark display filters to isolate DNP3 traffic by function code and event class.

REQUIRED SOFTWARE

- **Triangle MicroWorks DNP3 Test Harness** (free trial, Windows) — software master for outstation commissioning and SBO exercises.
- **Wireshark** (free, cross-platform) — protocol capture for frame-level analysis and SAv5 handshake verification.